



# Integrated Evaluation of Macro- and Microvascular Function in Human Limbs Using Simultaneous Application of FMD and NIRS Techniques

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**PURPOSE:** Endothelial dysfunction is an early indicator of cardiovascular disease. Regular exercise improves vascular function, highlighting the need for accurate assessment of exercise-induced vascular adaptations. This study introduces a simultaneous evaluation method that combines flow-mediated dilation (FMD) using Doppler ultrasound and vascular occlusion testing (VOT) using near-infrared spectroscopy (NIRS) to assess vascular responses at both the macrovascular and microvascular levels.

**METHODS:** We present a unified assessment protocol that integrates Doppler ultrasound-based FMD and NIRS-VOT. This study systematically outlines the experimental procedures, including participant preparation, equipment configuration, measurement protocols, data analysis methods, and clinical interpretation strategies. To ensure accurate and reproducible vascular-function assessment, standardized guidelines are provided for subject preparation (e.g., fasting and medication restriction), ultrasound imaging, NIRS probe placement, and occlusion parameters such as pressure and duration.

**RESULTS:** This combined approach provides a non-invasive means to assess vascular function at both the macrovascular and microvascular levels while also capturing their functional interactions.

**CONCLUSIONS:** The proposed method enables an integrated assessment of vascular adaptations across multiple levels, offering insights into the physiological mechanisms underlying exercise-induced vascular improvements. It may serve as a valuable tool in exercise research and clinical evaluation of vascular health.

**Key words:** Vascular function assessment, Flow-mediated dilation, Near-infrared spectroscopy, Reactive hyperemia

## INTRODUCTION

Vascular endothelial cells play a crucial role in regulating vascular tone and blood flow through the synthesis and secretion of various vasoactive substances [1]. Endothelial dysfunction is the earliest detectable indicator for predicting cardiovascular disease. Since endothelial dysfunction precedes morphological and clinical symptoms, assessing endothelial function enables early identification of cardiovascular disease risk [2]. Flow-mediated dilation (FMD) is a non-invasive method for

measuring endothelial function. When shear stress on the vessel wall increases, endothelial cells secrete nitric oxide (NO), a vasodilatory substance. FMD involves restricting blood flow for a predefined period, followed by inducing a hyperemic response upon cuff release to elevate shear stress on the vascular wall, thereby assessing the resulting NO-dependent vasodilatory response [3]. In contrast, vasodilation at the microvascular level occurs predominantly via NO-independent pathways. In human microvessels, the hyperemic response following transient occlusion is primarily mediated by the activation of inwardly rectifying potas-

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sium (KIR) channels and  $\text{Na}^+/\text{K}^+$ -ATPase, which together induce vascular smooth muscle hyperpolarization and subsequent vasodilation [4].

Regular exercise induces vascular adaptation in both conduit arteries and at the microvascular level [5-7]. While FMD is useful for assessing vascular function at the conduit artery level, it has limitations in observing microvascular responses where actual oxygen exchange occurs. Therefore, additional vascular function assessment is needed to complement FMD. For this purpose, many recent studies are using near-infrared spectroscopy (NIRS) equipment alongside FMD to evaluate microvascular responses [8,9]. NIRS primarily reflects microvascular function, as the detected signals originate from small vessels—such as arterioles, capillaries, and venules—while light is almost completely absorbed in vessels larger than 1 mm in diameter [10]. Hemoglobin (Hb) and myoglobin (Mb) each contain an iron-based heme group whose light absorption characteristics in the near-infrared spectrum vary depending on oxygen binding. NIRS leverages these spectral differences to quantify concentrations of oxygenated, deoxygenated, and total [Hb+Mb], as well as tissue oxygen saturation, defined as the ratio of oxy- to total [Hb+Mb] [11]. NIRS-derived metrics such as desaturation and reperfusion slopes can distinguish between age groups [12], training status [8], and cardiovascular risk profiles [13]. In particular, NIRS-based vascular occlusion test (VOT) is widely used as a strategy for assessing microvascular reactivity and can be conducted simultaneously with FMD because the two procedures share methodological similarities. The integrated use of FMD and NIRS-VOT enables not only individual assessment of vascular responses at each level but also comprehensive evaluation of macro-microvascular interactions by examining how downstream dilation affects upstream shear stress and the resulting endothelium-dependent vasodilation of conduit arteries.

The simultaneous assessment of FMD and NIRS-VOT will serve as a valuable tool for understanding exercise-induced improvements in vascular health. This technical paper aims to systematically organize the methodological and analytical considerations for this experimental approach.

## METHOD DEVELOPMENT

### 1. Participant preparation

To ensure accurate and reliable vascular function assessment, consistent subject preparation is required. This includes control of vitamin and antioxidant intake, medications, exercise, caffeine and alcohol consumption,

tobacco use, food intake, and menstrual phase. Conducting the assessment in the early morning hours, soon after waking, can help participants naturally comply with most of the preparation requirements. Additionally, vitamin and certain medication restrictions require adherence beginning two to three days prior to testing. Therefore, it is essential to inform these requirements to participants in advance. Recommendations outlined in this section are primarily derived from Harris et al. [14] and Thijssen et al. [3], which are the most widely cited FMD guidelines.

#### 1) Vitamin and antioxidant supplementation

Participants should avoid vitamin and antioxidant supplementation for at least 12 hours [3], and preferably 72 hours [14], before FMD assessment to prevent potential influences on endothelial function. Vitamins such as C, E, and  $\alpha$ -lipoic acid are known to modulate circulating free radicals and thereby impact FMD outcomes [3,14].

#### 2) Medications

Participants should abstain from medications that may influence vascular function for approximately four half-lives prior to testing [3]. This is particularly important for cardiovascular medications (e.g.,  $\beta$ -blockers, nitrates, calcium channel blockers). Non-steroidal anti-inflammatory drugs (NSAIDs) and aspirin should be discontinued at least 1 day and 3 days before testing, respectively [14].

If medication cessation is not feasible, the timing of the FMD assessment should be planned strategically to minimize potential drug effects. For example, measurements could be scheduled immediately prior to daily medication intake when plasma concentrations are expected to be at their lowest. Consistency in medication timing is essential, particularly for repeated measurements.

#### 3) Dietary restrictions

Participants should adhere to a fasted state for at least 6 hours prior to FMD assessment [3]. If fasting is impractical, only a low-fat meal is permitted [14], as high-fat and high-carbohydrate meals may impair FMD responses due to oxidative stress and hyperglycemia, whereas low-fat meals typically have negligible impact.

Additionally, participants should refrain from consuming caffeine, alcohol, and polyphenol-rich foods or beverages for at least 12 hours before testing [3,14], as these substances can modulate endothelial function and vasodilatory mechanisms.

#### 4) Tobacco use

Participants must avoid smoking and exposure to second-hand smoke for at least 6 hours [3], and preferably 12 hours [14], before testing, as tobacco negatively affects endothelial function.

#### 5) Exercise

Participants should refrain from moderate to vigorous physical exercise for at least 12 hours [14], and preferably 24 hours [3], before FMD assessment to prevent transient improvements in vascular function.

#### 6) Menstrual phase

For premenopausal women, testing should be consistently conducted during the early menstrual phase (days 1–7) [3,14], when estrogen and progesterone concentrations are at their lowest, to minimize hormonal variability in FMD responses.

#### 7) Acclimation phase

Participants should rest in a supine position for at least 10 to 15 minutes prior to FMD assessment. The testing environment should be maintained at a comfortable room temperature (e.g., 22–24°C). Measurements should be performed in a quiet, preferably darkened room to minimize sympathetic nervous system activity [3,14].

#### 8) Diurnal variation

Given the potential for diurnal variation in FMD responses, testing should be conducted at the same time of day across participants and repeated measures whenever possible. To enhance consistency between pre- and post-test conditions, the use of brief questionnaires to assess sleep quality and quantity is recommended. In particular, the Consensus Sleep Diary (CSD) offers a standardized and practical tool for capturing daily variations in sleep behavior [15].

## 2. Equipment preparation: ultrasound

### 1) Probe selection and frequency setting

It is recommended to use a linear ultrasound probe with a frequency of at least 7.5 MHz to optimize the resolution of B-mode images [3]. A trade-off exists between image resolution and penetration depth: as probe frequency increases, resolution improves but penetration depth decreases; conversely, lowering the frequency enhances penetration but reduces resolution [16]. Therefore, the probe frequency should be carefully adjusted to match the target vessel's depth and anatomical characteristics.

For example, Thijssen et al. [17] performed FMD measurements in the superficial femoral artery using a 7.5 MHz probe in healthy adults, including a group with an average body fat percentage of 29%. This indicates that a 7.5 MHz probe can provide sufficient penetration to image deeper vessels even in individuals with relatively high adiposity. Additionally, Schreuder et al. [18] have used 10 MHz probes to assess the superficial femoral artery in participants with lower BMI ( $\approx 22.8$ ), likely benefiting from improved resolution under conditions of reduced subcutaneous fat.

### 2) Duplex mode and related parameters

The primary stimulus for NO-dependent dilation is shear stress exerted on the vascular wall. To accurately calculate shear stress, simultaneous measurement of vessel diameter and blood velocity using duplex mode is essential.

Blood flow velocity is lower near the vessel wall and higher at the center of the vessel. Therefore, using a small sample volume that measures only the central flow velocity may lead to an overestimation of volume flow [14]. To ensure accuracy, the sample volume should be set as wide as possible within the vessel lumen without encroaching on the vessel walls. Additionally, it is advisable to allow a slight margin to account for potential movement during measurement. The same sample volume should be used consistently across repeated measurements.

While a 90° insonation angle provides optimal B-mode imaging, it is not suitable for measuring blood flow velocity. Thus, adjustment of the insonation angle is required to measure artery diameter and flow velocity simultaneously. Given that velocity measurement errors significantly increase at angles greater than 60°, an insonation angle of 60° or less is recommended [19]. In addition, the insonation angle should be consistently maintained across repeated pre- and post-intervention measurements within the same subject.

Historically, electrocardiogram (ECG) gating was utilized to measure vessel diameter at end-diastole in order to minimize the influence of cardiac cycle-related diameter fluctuations on FMD outcomes. However, recent studies have demonstrated that calculating the mean diameter across the entire cardiac cycle yields results that are highly consistent with those obtained using ECG gating [20]. Therefore, ECG gating is no longer considered an essential requirement for this assessment. Furthermore, continuous analysis of vessel diameter throughout the cardiac cycle without ECG gating simplifies equipment requirements and reduces analysis time.

### 3. Equipment preparation: NIRS

#### 1) Types of NIRS devices

NIRS is a technique that non-invasively measures the oxygenation status of hemoglobin and myoglobin in biological tissues. This method exploits the differences in light absorption between oxygenated and deoxygenated forms of these chromophores. NIRS uses light in the range of approximately 650–900 nm, which penetrates deeper into tissue than visible light. This property allows for the assessment of muscle oxygenation status in vivo [11].

NIRS devices are classified into three types according to the method by which light is delivered and detected: continuous wave (CW), time domain (TD), and frequency domain (FD). Among these, CW NIRS is the most commonly used due to its low cost and high portability. Hendrick et al. [21] conducted a review analyzing 31 studies over the past 30 years that utilized NIRS to assess post-occlusive reactive hyperemia. Excluding one study that did not specify the type of NIRS used, all studies employed either CW (23 studies) or FD (7 studies) NIRS.

While CW NIRS has been widely utilized, it has the limitation of being unable to measure the absolute concentration of the heme component [11]. To measure the absolute concentration, it is necessary to calculate the actual path length of the light, which is affected by the scattering coefficient. CW NIRS operates on the assumption that the scattering coefficient is constant. However, in reality, the scattering coefficient varies not only between subjects but also within the same subject between rest and exercise [22], thereby violating this assumption of CW NIRS. Therefore, while CW NIRS is suitable for tracking relative changes in tissue oxygenation, it is limited in its ability to measure absolute concentrations. Nevertheless, in VOT studies, the primary analytical endpoints are based on relative changes in tissue oxygenation [23], so this limitation of CW NIRS does not significantly compromise experimental validity.

#### 2) NIRS probe placement and site preparation

The placement of the NIRS probe should be determined by considering the study design and the characteristics of the participants. In studies involving exercise interventions, the muscle group engaged during exercise should be selected [21]. For studies involving obese participants, it is advisable to select a site with relatively low adipose tissue thickness (ATT) [21]. The penetration depth of the NIRS signal generally corresponds to approximately half the source-detector distance of the NIRS device [11]. When using a device with adjustable source-detector dis-

tance, increasing this distance allows for greater sampling depth but may reduce signal strength [21]. The specific experimental procedures are as follows:

- (1) ATT Measurement: Measure the ATT at the intended measurement site using a skinfold caliper or ultrasound to verify that it does not exceed half of the source-detector distance of the NIRS device. This ensures that the NIRS signal can adequately penetrate to the target muscle layer. The calf muscle typically has a relatively thin ATT compared to other lower limb regions [24] and is actively engaged in a variety of exercises. Therefore, it is a suitable site for NIRS probe placement in exercise physiology research.
- (2) Hair Removal: Remove any hair at the NIRS attachment site if present.
- (3) Location Marking: To ensure within-subject consistency, record the probe placement location using a specific anatomical landmark. For example, when measuring the medial head of the gastrocnemius, record the location as 20 cm vertically and 5 cm horizontally from the medial malleolus.
- (4) Probe Fixation and Light Blocking: After attaching the NIRS probe, secure it with black tape or a cover to prevent light leakage. As NIRS is a light-based measurement device, care must be taken to prevent ambient light from affecting the signal.

### 4. Equipment preparation: rapid cuff inflator

#### 1) Cuff position

The pressure cuff should be placed distal to the ultrasound probe and proximal to the NIRS probe (Fig. 1, 2). While both FMD and NIRS-VOT aim to induce a hyperemic response and evaluate the resulting vasodilation, the mechanisms underlying vasodilation differ between these techniques.

FMD specifically evaluates NO-dependent vasodilation, often referred to as NO-mediated vasodilation. This is achieved by isolating the vasodilatory response attributed to NO secreted from the vascular endothelium. To achieve this objective, the occlusion cuff is placed distal to the probe to avoid confounding factors. When the cuff is positioned proximal to the probe, the measurement site may become hypoxic. This hypoxic stimulus complicates the interpretation of the vasodilatory response observed after cuff release, as it becomes challenging to distinguish whether the response is due to increased shear stress leading to NO release or to the preceding hypoxic condition. Supporting this concept, previous studies have shown that when the cuff is placed proximal

to the probe, vasodilation persists even after the administration of NO blockers [24]. These findings indicate that placing the cuff distal to the probe is essential for accurately assessing endothelium-dependent vasodilation in FMD studies. In contrast, NIRS-VOT aims to evaluate hypoxia-induced vasodilation and therefore places the probe proximal to the occlusion site.

## 2) Occlusion duration

The recommended occlusion time for FMD is 5 minutes [3]. Shorter occlusion times may fail to elicit a sufficient vasodilatory response, while longer occlusion times may induce a more pronounced response but increase participant burden and potentially alter the underlying mechanism of vasodilation.

In NIRS-VOT, occlusion time can be set using two approaches: a fixed-time approach and a hypoxic stimulus-matched approach [25]. The fixed-time approach is identical to the method used in FMD assessments, applying the same occlusion time to all participants. Occlusion durations of 3 minutes and 5 minutes are most commonly employed [21]. However, the fixed-time approach has a limitation because the hypoxic stimulus can vary depending on the participant's basal skeletal muscle metabolic rate. Even with the same occlusion time, participants with higher metabolic rates may experience a greater hypoxic stimulus [12]. Studies comparing young adults and older adults have shown that young adults reach lower tissue oxygen saturation ( $\text{StO}_2$ ) levels [12]. Similar results have been observed when comparing athletes and non-athletes, with athletes reaching lower  $\text{StO}_2$  levels than their non-athlete counterparts [8].

Since the hypoxic stimulus is the primary driver of microvascular vasodilation assessed by NIRS-VOT, inter-individual differences in the hypoxic stimulus can complicate the interpretation of study results. To address this limitation, a hypoxic stimulus-matched approach has been developed, in which a pre-specified desaturation level is defined, and the cuff is released once that level is reached. Although this approach allows for the standardization of the hypoxic stimulus across participants in NIRS-VOT assessments, it is not compatible with simultaneous FMD assessment. This is because FMD protocols utilize a fixed-time occlusion approach, whereas hypoxic stimulus-matched approach in NIRS-VOT inevitably leads to variable occlusion durations across participants, depending on their individual metabolic rates.

Therefore, it is recommended to select one of the following approaches based on the study objectives and design considerations. (1) Employ a

fixed-time approach to conduct simultaneous FMD and NIRS-VOT assessments. An occlusion time of 5 minutes is recommended, as it is the most commonly used duration in both FMD and NIRS studies. (2) Conduct separate FMD and NIRS-VOT assessments. If a sufficient recovery period (e.g., 20 minutes) is ensured between tests, comparable results can be obtained using either simultaneous or separate measurements [9]. Although measurement timing can critically affect outcomes in acute response studies, a 20-minute difference in measurement timing is unlikely to significantly impact results in studies examining vascular adaptations following regular training. Therefore, in such cases, it may be a reasonable alternative to perform separate measurements, using a fixed-time approach for FMD and a hypoxic stimulus-matched approach for NIRS-VOT.

## 3) Occlusion pressure

A cuff pressure of 250 mmHg has traditionally been used, but a recently published NIRS-VOT protocol [23] recommends applying a lower occlusion pressure. This recommendation is due to the risk of participant withdrawal caused by discomfort and pain associated with high occlusion pressures. Kriel et al. [23] recommended an occlusion pressure of 200 mmHg, while our laboratory protocol applies 220 mmHg. In general, a pressure at least 50 mmHg above the participant's systolic blood pressure is considered sufficient to achieve arterial occlusion. Therefore, unless the participant's blood pressure is particularly high, an occlusion pressure between 200 and 220 mmHg is likely adequate. Kriel et al. [23] verified complete occlusion using strain gauge plethysmography, whereas in our laboratory, complete occlusion at 220 mmHg was confirmed via ultrasound Doppler.



**Fig. 1.** Experimental set-up for simultaneous FMD and NIRS assessments. (A) Upper limb set-up. (B) Lower limb set-up.

#### 4) Participant eligibility

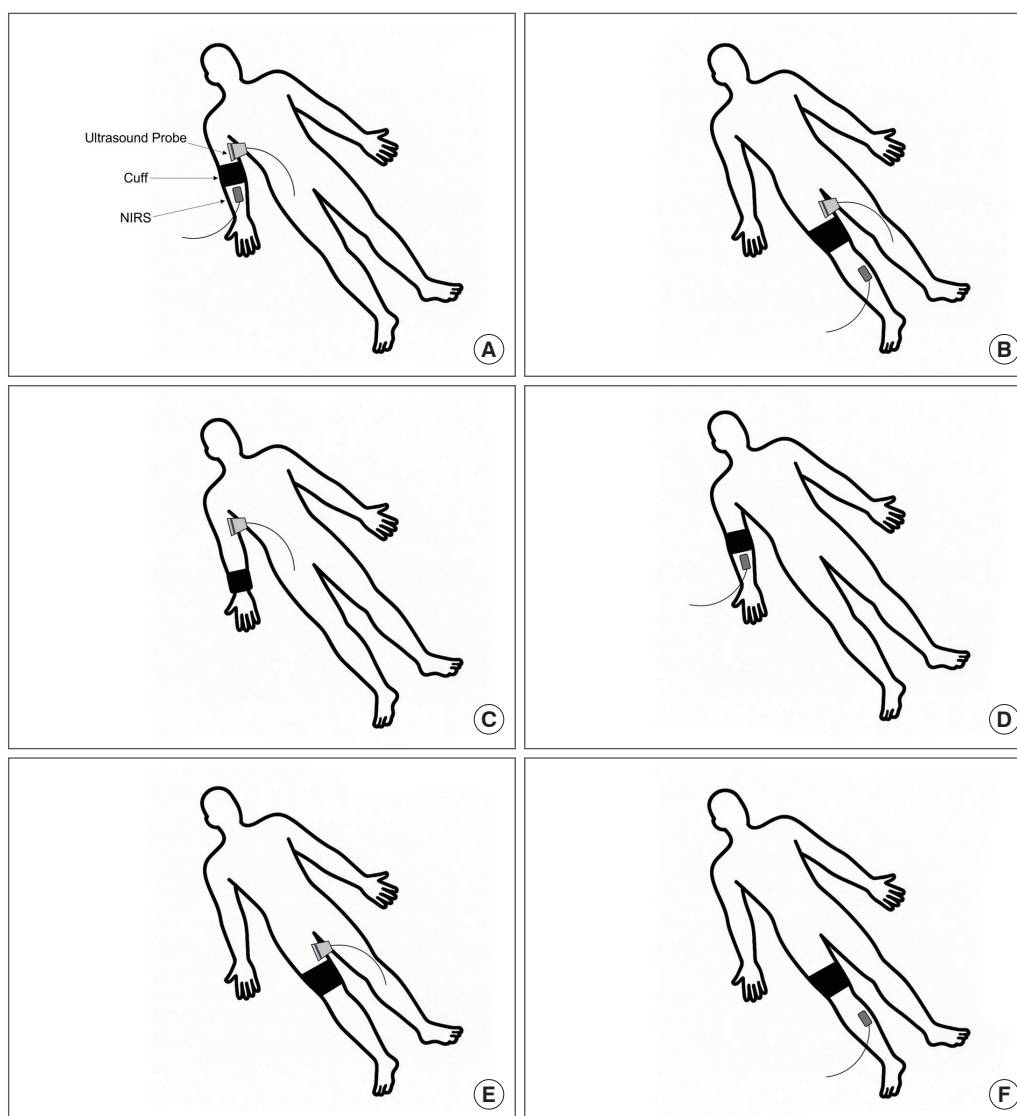
Vascular occlusion test in the lower limbs is contraindicated in individuals with a history of revascularization procedures involving vascular grafts or stenting of the femoral or popliteal arteries [23].

ATT should be assessed using skinfold calipers or ultrasound to ensure that it does not exceed half of the NIRS device's source-detector separation distance, as near-infrared light typically penetrates tissue to a depth of approximately half this distance [11].

#### 5. Experimental procedure

##### 1) Baseline measurement

Instruct the participant to rest in a supine position to ensure hemodynamic stabilization before beginning data collection. While the participant is resting, verify that the NIRS signal is being acquired properly. Record baseline arterial diameter and blood flow velocity continuously for at least 1 minute. Identify the segment with the clearest B-mode ultrasound image and use this portion for calculating the representative baseline diameter.



**Fig. 2.** Example of ultrasound probe, NIRS device, and occlusion cuff placement for simultaneous and non-simultaneous FMD and NIRS assessments. (A) Simultaneous assessment of FMD and NIRS in the upper limb. Ultrasound probe placed on the brachial artery, NIRS device placed on the forearm, and occlusion cuff placed on the elbow crease. (B) Simultaneous assessment of FMD and NIRS in the lower limb. Ultrasound probe placed on the superficial femoral artery, NIRS device placed on the gastrocnemius muscle, and occlusion cuff placed on the distal thigh. (C, D) Non-simultaneous measurements in the upper limb. (C) Ultrasound probe placed on the brachial artery and occlusion cuff placed on the wrist. (D) NIRS device placed on the forearm and occlusion cuff placed on the elbow crease. (E, F) Non-simultaneous measurements in the lower limb. (E) Ultrasound probe placed on the superficial femoral artery and occlusion cuff placed on the distal thigh. (F) NIRS device placed on the gastrocnemius muscle and occlusion cuff placed on the distal thigh.

## 2) Occlusion period

Set the occlusion pressure of the rapid cuff inflator to at least 50 mmHg above the participant's systolic blood pressure. As NIRS data are continuously recorded throughout the procedure, mark the time point at the start of occlusion for later analysis. Inform the participant before the occlusion begins, and instruct them to remain as still as possible. Mark the probe placement site used for baseline data collection to ensure consistent measurement locations. This allows post-deflation measurements to be obtained from the same region, even if the participant moves during the procedure. Leave a small space under the cuff to minimize displacement of the measurement site during inflation and deflation. Begin ultrasound data acquisition 1 minute prior to cuff release.

## 3) Post-deflation measurement

Accurate data collection during the post-deflation period is critical, as several key variables for both FMD and NIRS-VOT are measured during this phase. Before cuff release, inform the participant that they may experience tingling or discomfort during reperfusion and instruct them to remain as still as possible. This precaution minimizes motion artifacts and ensures data quality. After cuff release, continue data acquisition for at least 3 minutes for the upper limb and at least 5 minutes for the lower limb. Record the ultrasound probe and NIRS device placement sites to maintain consistency in repeated-measures designs.

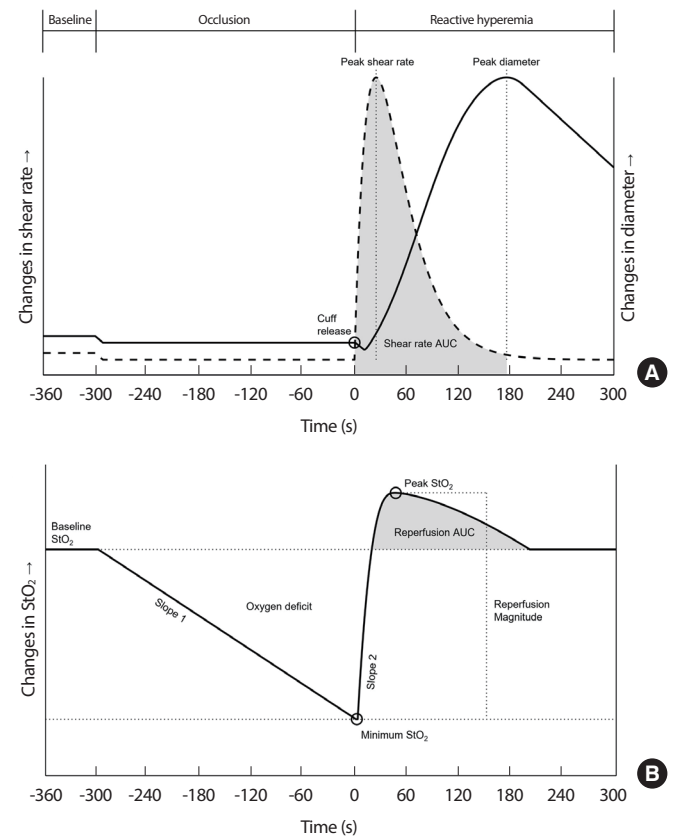
## 6. Analytical endpoints (FMD)

The key analytical endpoints of the FMD assessment are summarized in Table 1 and Fig. 3A. Accurate assessment of FMD requires reliable interpretation of vascular ultrasound images, making the selection of an

analytic method a critical component of the overall protocol. FMD data analysis can be broadly categorized into manual and automated approaches. Manual analysis is highly operator-dependent, time-consuming, and susceptible to observer bias, which may compromise reproducibility [3]. To address these limitations, automated software equipped with wall-tracking algorithms has been increasingly adopted. These systems significantly reduce analysis time and minimize user-related error [3].

Brachial Analyzer (MIA LLC, IA, USA) is the most widely used automated software for FMD analysis. In our laboratory, we employed FMD Studio (Institute of Clinical Physiology, Pisa, Italy), a validated software that has demonstrated strong agreement with Brachial Analyzer [26]. Notably, FMD Studio does not require ECG tracking and supports real-time processing, enabling simultaneous image acquisition and analysis.

**Baseline Diameter:** The baseline diameter refers to the resting diameter of the artery, reflecting the structural characteristics of the vessel. It serves as the reference point for calculating FMD. The baseline diameter



**Fig. 3.** Schematic presentation of vascular and hemodynamic responses before, during, and after a 5-minute ischemic stimulus. (A) Diameter and shear rate responses at macrovascular level during FMD assessment. (B) StO<sub>2</sub> responses at microvascular level during NIRS-VOT.

**Table 1.** Flow-mediated dilation analytical endpoints: measurement and calculation

| Variable               | Assessment / Calculation                                                                                                          |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Baseline Diameter (mm) | Mean diameter of $\geq 10$ cardiac cycles pre-occlusion                                                                           |
| Peak Diameter (mm)     | Continuous ultrasound tracking post-occlusion                                                                                     |
| FMD (mm)               | Peak Diameter–Baseline Diameter                                                                                                   |
| FMD (%)                | Percentage change from baseline;<br>$\frac{\text{Peak Diameter} - \text{Baseline Diameter}}{\text{Baseline Diameter}} \times 100$ |
| Blood Flow (mL/s)      | $\text{Velocity} \times \pi \left( \frac{\text{Diameter}}{2} \right)^2$                                                           |
| Shear Rate (1/s)       | $\frac{4 \times \text{Velocity}}{\text{Diameter}}$                                                                                |

is determined by averaging the artery's diameter over 10 cardiac cycles or during a 30-second period before cuff inflation.

**Peak Diameter:** Peak diameter denotes the maximum arterial diameter observed following cuff deflation, representing the vessel's maximal vasodilatory response. This is defined as the highest diameter value averaged over a 3–10 seconds period after cuff deflation. A time bin that is too short may introduce excessive noise, while one that is too long may smooth the data excessively, potentially underestimating the true peak diameter. Rodriguez-Miguel et al. [27] recommends using a 5-second averaging window to balance the trade-off between noise and smoothing. In the upper limb, the peak diameter generally occurs between 45 and 80 seconds after cuff deflation [14]. However, in the lower limb artery, the peak response is delayed [3]. Accordingly, Thijssen et al. [3] recommends extending the post-deflation monitoring period beyond the conventional 3-minute window used in upper limb assessments when conducting FMD assessment in the lower limb.

**FMD:** NO-dependent vasodilation in response to increased shear stress is commonly reported in two forms: absolute and percent change in arterial diameter. Absolute FMD is calculated as the difference between the peak and baseline arterial diameters and is expressed in millimeters. Percent FMD (%FMD), is computed by dividing the absolute change by the baseline diameter and multiplying by 100.

**Blood Flow:** Blood flow refers to the volume of blood passing through a vessel per unit time. It is calculated by multiplying the cross-sectional area of the artery by the mean blood velocity. Analyzing the increase in blood flow during FMD assessment provides important complementary insight to %FMD interpretation. While %FMD is a valuable indicator of endothelial function, it may not fully capture vascular adaptations in individuals who have undergone long-term training. In these individuals, structural remodeling such as an enlarged resting arterial diameter facilitates increased blood delivery with reduced dependence on functional vasodilation [8]. This phenomenon, often referred to as the “athlete paradox,” describes the observation that trained individuals may show similar %FMD responses compared to untrained individuals, despite demonstrating a significantly greater hyperemic response after cuff release. This discrepancy arises because trained arteries are structurally adapted for higher flow capacity, allowing for greater absolute perfusion even with modest relative vasodilation. Therefore, blood flow assessment serves as a complementary tool to %FMD, especially when evaluating populations with substantial structural vascular adaptations.

**Shear Rate:** Shear stress is the primary physiological stimulus for NO-

mediated vasodilation. To evaluate the relationship between the mechanical stimulus and vascular response during reactive hyperemia, it is essential to quantify the shear stimulus applied to the vessel wall. Shear rate serves as a surrogate marker of shear stimulus and is calculated by multiplying four times the mean blood velocity divided by the vessel diameter.

**Normalization of FMD:** Normalization strategies have been proposed to account for variables that influence %FMD, particularly shear stimulus and baseline arterial diameter. A commonly used approach involves dividing %FMD by shear rate-derived metrics to adjust for interindividual differences in the vasodilatory stimulus. This includes normalization by peak shear rate or shear rate area under the curve (AUC), with the latter calculated using the trapezoidal rule, as described in the *Reperfusion AUC* section below, either until peak dilation or over a fixed duration (e.g., 2 minutes). Such methods aim to refine the interpretation of endothelial function by aligning the vascular response with the magnitude of the applied stimulus. Baseline diameter is another critical factor affecting %FMD. Even when the absolute change in diameter is identical, individuals with smaller baseline diameters will yield higher %FMD values. For example, a 0.3 mm dilation corresponds to a 10% increase in a 3 mm artery but only 5% in a 6 mm artery. To address this mathematical bias, Atkinson & Batterham [28] proposed the use of allometric scaling, which adjusts for variations in baseline vessel size. Although several normalization techniques have been introduced, no consensus has been reached regarding the most appropriate method.

## 7. Analytical endpoints (NIRS–VOT)

The key analytical endpoints of the NIRS-VOT are summarized in Table 2 and Fig. 3B.

**Baseline StO<sub>2</sub>:** Baseline StO<sub>2</sub> represents the muscle oxygen saturation at rest and is calculated as the average over 1–2 minutes prior to cuff inflation. It serves as a reference point for quantifying the degree of ischemic stimulus during occlusion and the extent of hyperemic response following cuff release.

**Slope 1:** Slope 1 refers to the rate of decline in StO<sub>2</sub> following the onset of vascular occlusion. It reflects the local metabolic demand of skeletal muscle, with steeper slopes indicating higher oxygen consumption. This parameter serves as an indicator of the resting metabolic rate of skeletal muscle, and tends to decline with advancing age [12,29], being generally lower in untrained individuals compared to trained counterparts [8]. It is calculated over a defined interval, such as 30 to 150 seconds after cuff

**Table 2.** NIRS-VOT analytical endpoints: measurement and calculation

| Variable                      | Assessment / Calculation                                                                                                                                                                 |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline StO <sub>2</sub> (%) | Average StO <sub>2</sub> over 1–2 minutes prior to cuff inflation                                                                                                                        |
| Slope 1 (%/s)                 | Linear rate of StO <sub>2</sub> decline during occlusion (e.g., 30–150 s);<br>$\frac{(\Delta \text{StO}_2)}{\Delta t}$                                                                   |
| Minimum StO <sub>2</sub> (%)  | Lowest StO <sub>2</sub> value during occlusion                                                                                                                                           |
| Peak StO <sub>2</sub> (%)     | Highest StO <sub>2</sub> value after cuff release                                                                                                                                        |
| Reperfusion Magnitude (%)     | Difference between Peak and Minimum StO <sub>2</sub> values;<br>Peak StO <sub>2</sub> – Minimum StO <sub>2</sub>                                                                         |
| O <sub>2</sub> deficit (%·s)  | Area between the baseline StO <sub>2</sub> and the desaturation curve during occlusion; Calculated using the trapezoidal rule (see <i>Reperfusion AUC</i> section)                       |
| Slope 2 %/s                   | Linear rate of StO <sub>2</sub> increase during the first 10 seconds post-deflation; calculated using the same formula as Slope 1                                                        |
| Reperfusion AUC (%·s)         | Area under the StO <sub>2</sub> curve above baseline following cuff release, assessed over a 2-minute period; Calculated using the trapezoidal rule (see <i>Reperfusion AUC</i> section) |

inflation, using linear regression [30]. The value is expressed in percentage per second (%/s).

*Minimum StO<sub>2</sub>, Peak StO<sub>2</sub>, and Reperfusion Magnitude:* Minimum StO<sub>2</sub> refers to the lowest level of muscle oxygen saturation observed during the occlusion phase and reflects the severity of ischemic stress applied to the tissue. Peak StO<sub>2</sub> represents the highest oxygen saturation attained after cuff release, typically occurring during the reactive hyperemia phase. The reperfusion magnitude, defined as the difference between Peak StO<sub>2</sub> and Minimum StO<sub>2</sub>, quantifies the amplitude of the microvascular recovery response.

*O<sub>2</sub> deficit:* O<sub>2</sub> deficit is calculated as the area between the baseline StO<sub>2</sub> and the declining StO<sub>2</sub> curve during the occlusion period. The integration window spans from the onset of cuff inflation to the time of cuff release. The upper boundary is defined by the baseline StO<sub>2</sub>, while the lower boundary is formed by the ischemia-induced decline in StO<sub>2</sub>. The area is enclosed by these boundaries and a vertical line drawn at the time of deflation.

*Slope 2:* Slope 2 reflects the initial reperfusion rate following cuff release and serves as an indicator of microvascular reactivity during the early phase of reactive hyperemia. Slope 2 shows a strong association with FMD and reflects the vascular responsiveness of the microvasculature [30]. During the ischemic period, downstream accumulation of metabolic by-products induces microvessel dilation, which contributes to a robust hyperemic response following cuff release and stimulates upstream conduit artery dilation [30]. This parameter is typically greater in lean than obese individuals [31], in healthy controls compared with sepsis patients—where lower values are linked to greater disease severity [32]—and in trained compared to untrained subjects, with significant

correlations observed with maximal oxygen uptake [8]. It is calculated as the slope of the StO<sub>2</sub> increase within the first 10 seconds post-deflation, expressed in percentage per second (%/s). Extending the analysis window may reduce the linearity of the response, as the signal may reach its peak and begin returning toward baseline, thereby incorporating non-linear segments [9]. Alternatively, some studies report the time required to reach 50% of the reperfusion magnitude (half-time) as an index of recovery kinetics [33].

*Reperfusion AUC:* Reperfusion AUC represents the integrated hyperemic response following cuff release and captures both the magnitude and duration of tissue reoxygenation. Reperfusion AUC has been reported to distinguish variations in microvascular responsiveness according to body composition and age [31,34]. Higher values are observed in lean compared to obese individuals [31] and in young compared to middle-aged and older adults [34]. Moreover, AUC values are inversely associated with arterial stiffness, a well-established indicator of atherosclerosis [34]. It is calculated as the area under the StO<sub>2</sub> curve that lies above the baseline level, with the baseline StO<sub>2</sub> serving as the lower boundary. The analysis is typically performed over a 2-minute period following cuff deflation [9]. AUC is calculated using the trapezoidal rule, according to the following equation.

$$AUC = \sum_{i=1}^n \frac{(y_i + y_{i+1})}{2} (x_{i+1} - x_i)$$

Shear rate AUC for FMD normalization, as well as O<sub>2</sub> deficit and reperfusion AUC in NIRS-VOT, are all calculated using the trapezoidal integration method described above.

## VALIDATION

FMD is a well-established, noninvasive index of endothelial function. It is impaired in individuals with cardiovascular risk factors and improves with effective interventions, supporting its criterion validity [35]. Pharmacological blockade studies [36–38] further confirm that FMD primarily reflects NO-dependent vasodilation, establishing it as a validated tool for macrovascular assessment. At the microvascular level, NIRS-VOT demonstrates convergent validity by differentiating post-occlusive hyperemia and reoxygenation kinetics across diverse clinical and demographic groups, including cardiovascular and metabolic diseases, respiratory disorders, smoking status, and aging cohorts [21]. The reproducibility of key NIRS-VOT parameters has been demonstrated, with intraclass correlation coefficients ranging from moderate to excellent (0.62–0.84) and inter-observer reliability reported as excellent to perfect (ICC 0.85–1.00) [39]. The high reproducibility of NIRS-VOT supports its use for repeated measurements, making it a valuable tool in exercise physiology research, as it provides a reliable means to detect acute vascular changes following exercise as well as long-term adaptations to training. Previous work further demonstrated that the association between NIRS-derived indices and FMD is preserved under both simultaneous ( $r=0.60$ ) and non-simultaneous ( $r=0.62$ ) testing approaches [9], supporting the feasibility of concurrent assessment. Importantly, integrating FMD and NIRS-VOT enhances sensitivity to physiological differences and captures group variations that might be overlooked when relying on either method alone [8]. Together, these techniques provide validated macro- and microvascular endpoints suitable for comprehensive vascular assessment.

## DISCUSSION

Macro- and microvascular assessments are complementary and may be advantageously combined when evaluating vascular function. Conduit arteries serve as primary pathways for blood flow, whereas microvessels represent the sites of tissue perfusion. FMD provides an index of conduit artery endothelial function, while NIRS-VOT captures microvascular reactivity. Combining these methods allows researchers not only to examine vascular responses at each level independently but also to explore potential interactions between macro- and microvascular function. This integration enhances both the breadth and depth of data interpretation.

FMD is a clinically validated and widely used index of endothelial function, with established associations with various cardiovascular diseases [40–42]. However, relying solely on FMD may be insufficient to fully capture vascular adaptations to exercise. The concept of the athlete's paradox illustrates this limitation, highlighting the need for complementary measures to comprehensively evaluate exercise-induced vascular changes. The athlete's paradox refers to the observation that athletes—who are expected to have superior vascular health—often show FMD values comparable to those of untrained individuals [43]. This occurs because FMD quantifies relative rather than absolute dilation by comparing vascular diameters during reactive hyperemia with the baseline diameter. Consequently, individuals with larger baseline diameters may exhibit lower FMD percentages even when the absolute increase in diameter is identical. With long-term exercise training, structural remodeling of the artery frequently increases baseline diameter and enhances its blood-conducting capacity, yet the relative FMD value may remain unchanged or even decrease [43]. In this context, structural adaptations resulting from chronic training can mask early functional improvements in endothelial responsiveness, leading to an apparent null effect in FMD outcomes.

In contrast, microvascular responsiveness assessed by NIRS-VOT exhibits greater sensitivity to exercise-induced adaptations. Slope 2, an index of microvascular reactivity, is significantly higher in trained than in untrained individuals [8]. Moreover, this enhancement is more pronounced in the muscle groups predominantly engaged during habitual training. Soares et al. [7] reported larger effects in the legs of runners and cyclists compared with the arms. Taken together, combining FMD and NIRS-VOT offers a powerful analytical strategy in exercise physiology. FMD provides a clinically validated assessment of conduit artery endothelial function, whereas NIRS-based parameters demonstrate high sensitivity to exercise-induced vascular adaptations.

The practical value of combining FMD and NIRS-VOT lies in their procedural similarity. Because the two tests share core procedures, they can be implemented within a single session, reducing measurement time and minimizing participant burden. Such efficiency is advantageous in both research and clinical settings where participant compliance and resource optimization are critical.

Despite these advantages, methodological limitations must be considered when conducting simultaneous assessments. When the NIRS probe is attached to the forearm, the occlusion cuff must be placed at the elbow so that the probe remains distal to the occlusion site. Data collected un-

der this configuration should be compared cautiously with FMD protocols that employ wrist occlusion. The more proximal cuff placement exposes a larger volume of distal tissue to ischemia, which may augment the FMD response [44] and complicate direct comparisons between protocols.

Future efforts should aim to establish a consensus on optimal probe placement and occlusion strategies to enhance reproducibility and comparability across studies. Systematic evaluation of NIRS sensor positioning with respect to reproducibility, validity, and reliability is essential. Such a consensus would reduce methodological heterogeneity and strengthen the interpretability of findings across laboratories.

## CONCLUSIONS

To our knowledge, this is the first study to present a systematic protocol for the combined use of FMD and NIRS-VOT. The framework, which specifies participant preparation, measurement procedures, and analysis steps, is designed to support consistent data collection and strengthen comparability across studies. This approach enables integrated evaluation of conduit artery and microvascular function and offers practical value for exercise science. In particular, it can be applied to examine acute vascular responses to single exercise sessions and long-term adaptations to exercise training.

## CONFLICT OF INTEREST

The authors declare that they do not have conflict of interest.

## AUTHOR CONTRIBUTIONS

Conceptualization: TJ Kim, JH Kim\*; Data curation: TJ Kim, JH Kim, H Kang; Formal analysis: TJ Kim, JH Kim, H Kang; Funding acquisition: JH Kim\*; Methodology: TJ Kim, JH Kim, H Kang; Visualization: TJ Kim; Writing - original draft: TJ Kim, JH Kim\*; Writing - review & editing: TJ Kim, JH Kim, H Kang, JH Kim\*.

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